Customer Churn Analysis (Project)

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Role: Data Analyst (Personal Project)

Description: This project focuses on analyzing customer churn behavior in an e-commerce business. The objective is to identify patterns, trends, and key factors that lead customers to stop purchasing, and provide actionable insights to improve customer retention and drive business growth.

The analysis was performed on a comprehensive dataset containing over 1 million records, structured across four key tables:

- Customers – Customer demographics, account status, and engagement metrics.
- Orders – Transaction history, order frequency, and order value.
- Payments – Payment methods, transaction status, and trends.
- Reviews – Customer feedback and ratings.

Tools and Techniques:

SQL: Used for data extraction, cleaning, aggregation, and basic analytics.
Python: Used for data processing, feature engineering, exploratory data analysis (EDA), and advanced calculations.
Visualization Libraries: Matplotlib, Seaborn, and Plotly were used to create detailed and interactive charts, enabling clear insights into customer behavior, churn trends, and segmentation analysis.

Business Problem

The company observed that although customer churn was relatively low, some customers were still not returning to make repeat purchases. Even a small amount of churn can impact long-term revenue and customer lifetime value. Therefore, the company wanted to proactively analyze customer behavior and identify potential risks before churn increases.

The company aimed to understand:

- Why some customers are leaving the platform
- Which customers are at risk of churning
- What factors influence customer churn
- How to reduce churn and improve customer retention
- How to increase repeat purchases and long-term customer value

Objectives

The main objective of this project was to analyze customer behavior and identify key factors contributing to customer churn. The goal was to provide actionable insights that help the business improve customer retention and long-term growth.

The specific objectives of this analysis were:

- Calculate overall customer churn rate
- Identify active vs churned customers
- Analyze monthly churn trends over time
- Understand customer purchasing behavior
- Compare high-value and low-value customers
- Perform RFM segmentation to identify customer groups
- Analyze the impact of customer reviews on churn
- Evaluate payment method behavior and churn patterns
- Identify geographic regions with higher churn rates
- Provide actionable insights to reduce churn and improve retention

1. What is the total number of customers?

Total Customers

49,000

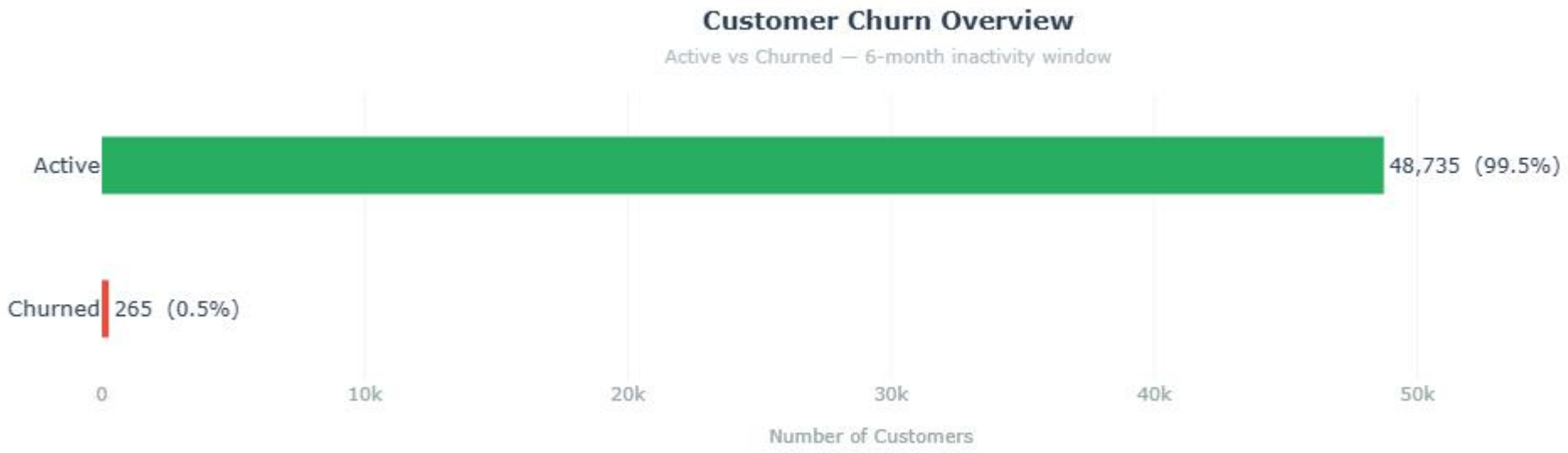
- Total customers analyzed: **49,000.**
- The dataset includes a moderate customer base for churn analysis.

2. How many customers are active vs churned?

- Total Active Customers: 48,735.
- Total Churned Customers: **265**.
- The majority of customers remain active, indicating strong overall retention.
- A small portion of customers have churned, but monitoring is still important to prevent future increase.

Actionable Insights :

1. Monitor churned customers to identify common churn patterns.
2. Focus on early detection of customers showing declining activity.
3. Implement retention strategies to maintain strong customer retention levels.



3. What is the overall churn rate?

- Overall churn rate: **0.54%.**
- The churn rate is very low, indicating strong customer retention.
- A small percentage of customers are leaving the platform.

Actionable Insights:

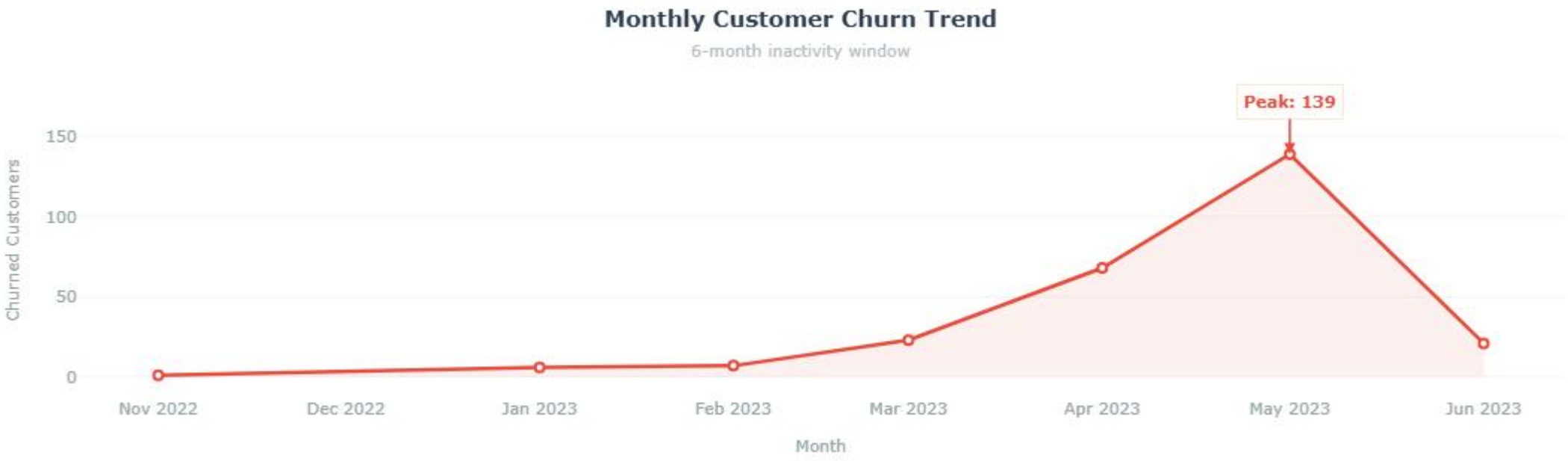
1. Maintain current customer engagement strategies to sustain low churn.
2. Identify churned customers to understand potential risk factors.
3. Monitor churn rate regularly to detect any future increase.

4. What is the monthly churn trend over time?

- Customer churn remained low during late 2022 and early 2023.
- Churn started increasing gradually from March 2023.
- A significant spike in churn was observed in May 2023.
- Churn declined sharply in June 2023 after the May peak.
- The churn pattern indicates a temporary surge rather than a consistent upward trend.

Actionable Insights:

1. Investigate business or operational changes during May 2023.
2. Identify customer segments contributing to the churn spike.
3. Implement proactive retention strategies during similar periods.
4. Continuously monitor monthly churn trends for early detection.



5. What is the average number of orders per customer?

- The average number of orders per customer is **20.41.**
- Customers show strong engagement with repeat purchases.
- A higher order frequency indicates strong customer retention and platform usage.

Avg Orders per Customer

20.41

Actionable Insights:

1. Identify high-frequency customers for loyalty and reward programs.
2. Encourage low-frequency customers to increase repeat purchases.
3. Use personalized recommendations to maintain high customer engagement.

6. What is the average customer spending and average order value?

- Average order value is **₹ 2,547**.
- Average customer spending is **₹ 51,990.**
- Customers show high overall spending across multiple purchases.
- Higher customer lifetime value indicates strong revenue contribution per customer.

Actionable Insights:

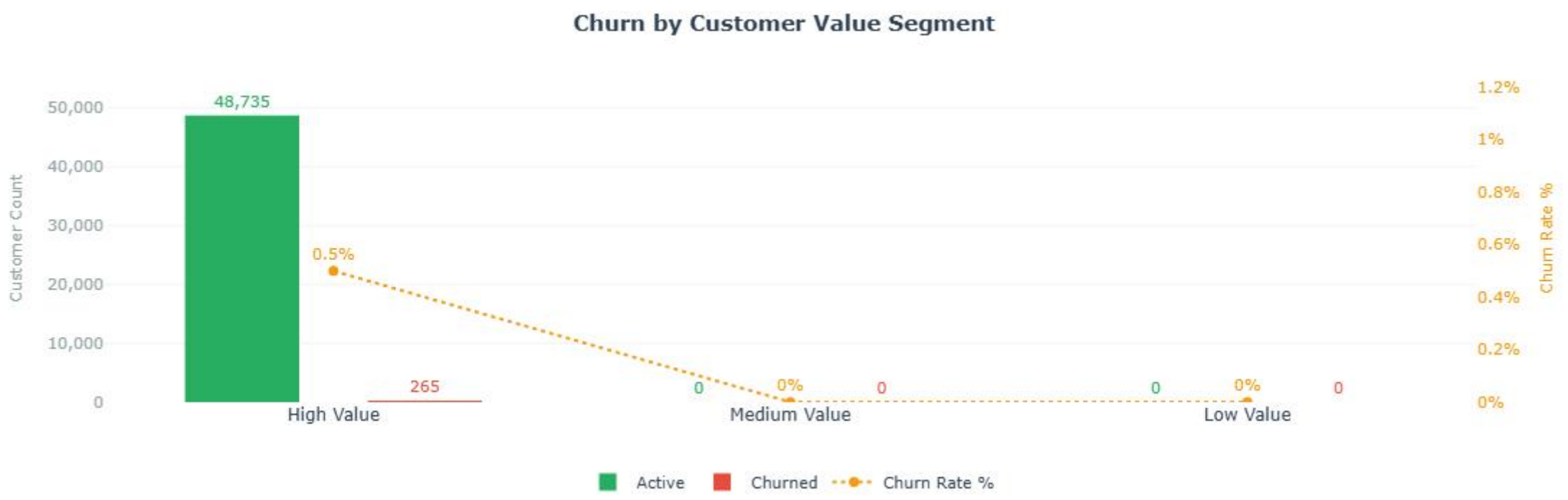
1. Focus on retaining high-spending customers.
2. Create loyalty programs for high-value customers.
3. Encourage upselling and cross-selling to increase average order value.

7. Comparison of high-value, Medium- value and low-value customers in terms of churn: High Value (≥2000), Medium Value (1000–1999), Low Value (<1000)

- All customers fall under the high-value segment (**≥ ₹2000 average order value**).
- Active high-value customers: 48,735.
- Churned high-value customers: 265.
- No customers were observed in the medium-value or low-value segments.
- High-value customers demonstrate strong retention with minimal churn.

Actionable Insights

- Continue focusing on retention strategies for high-value customers.
- Maintain strong customer experience to sustain customer loyalty.
- Monitor high-value customers showing declining engagement.

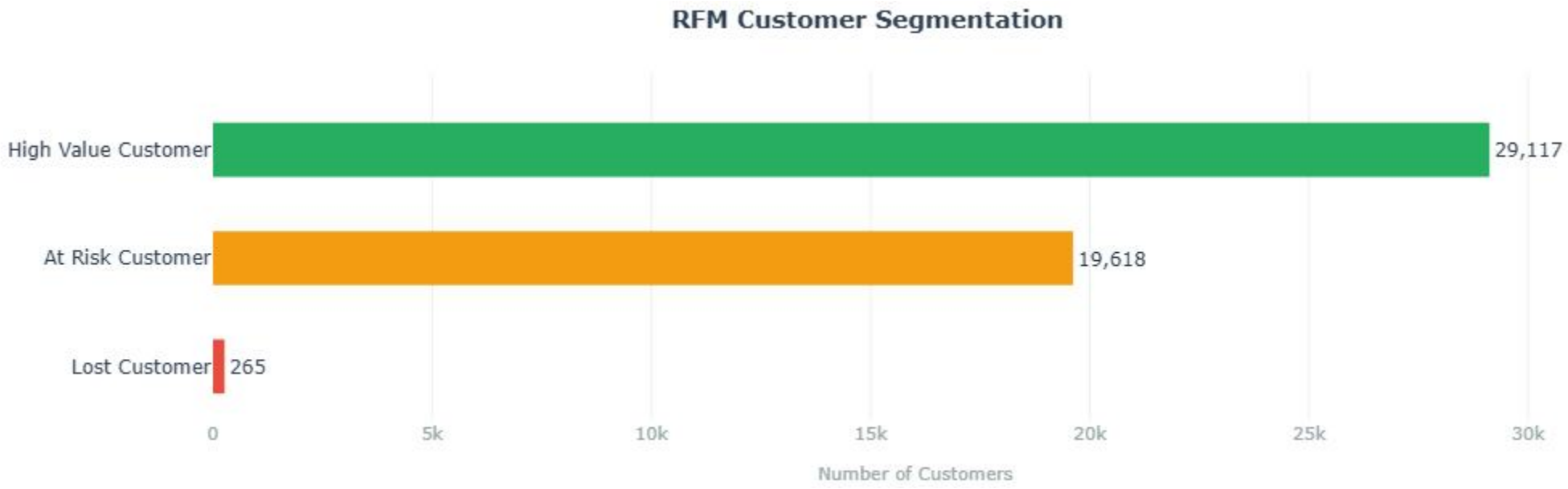

Churn by Customer Value Segment
Customer Count
50,000
40,000
30,000
20,000
10,000
0
Churn Rate %
1.2%
1%
0.8%
0.6%
0.4%
0.2%
0%
48,735
265
0.5%
0
0%
0
0
0%
0
High Value
Medium Value
Low Value
Active
Churned
Churn Rate %

8. Customers were segmented using RFM (Recency, Frequency, Monetary) analysis based on the following criteria:

- **Recency: Days since last purchase**
- **Frequency: Number of orders**
- **Monetary: Total customer spending**

Customer segments were defined as:

- **High-Value Customers → Recency ≤ 30 days AND Frequency ≥ 5 orders AND Monetary ≥ ₹2000**
- **At-Risk Customers → Recency > 30 days AND Recency ≤ 180 days**
- **Lost Customers → Recency > 180 days**



- High-value customers: 29,117.
- At-risk customers: 19,618.
- Lost customers: 265.
- A large portion of customers falls under the high-value segment, indicating strong customer engagement.
- A notable number of customers are categorized as at-risk, representing potential future churn.
- Lost customers account for a very small portion of the customer base.

Actionable Insights:

1. Focus on re-engagement strategies for at-risk customers.
2. Maintain loyalty programs for high-value customers.
3. Implement win-back campaigns for lost customers.

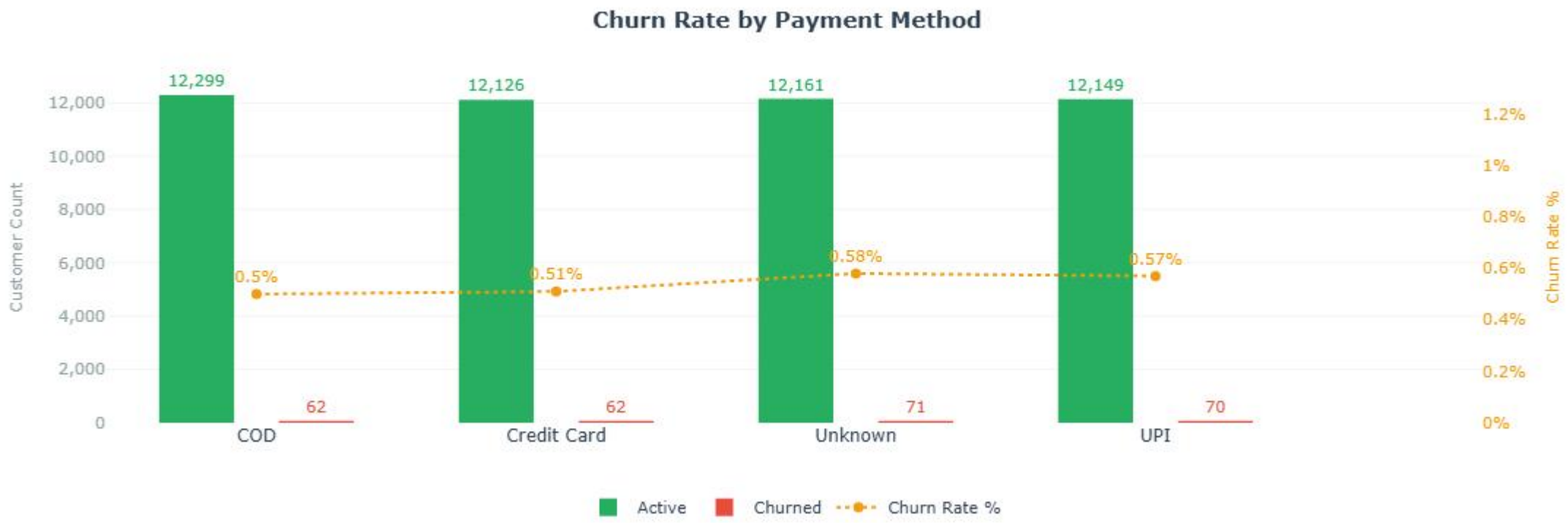
9. What is the impact of review scores on customer churn?

- Customers with review score **3 show similar numbers of active and churned customers.**
- Review scores at this level do not appear to strongly influence churn.
- Customer churn is low despite average review scores, indicating other factors drive retention.

Actionable Insights:

1. Monitor customers with low review scores to detect potential dissatisfaction.
2. Encourage feedback collection to improve product/service experience.
3. Focus retention strategies on engagement and purchase behavior rather than solely on review scores.

10. What is the churn rate by payment method (COD vs Online)?



- COD: 12,299 active, 62 churned.
- Credit Card: 12,126 active, 62 churned.
- UPI: 12,149 active, 70 churned.
- Unknown payment method: 12,161 active, 71 churned.
- Churn rates are very low across all payment methods, with minor differences.
- No single payment method shows a significant impact on churn.

Actionable Insights:

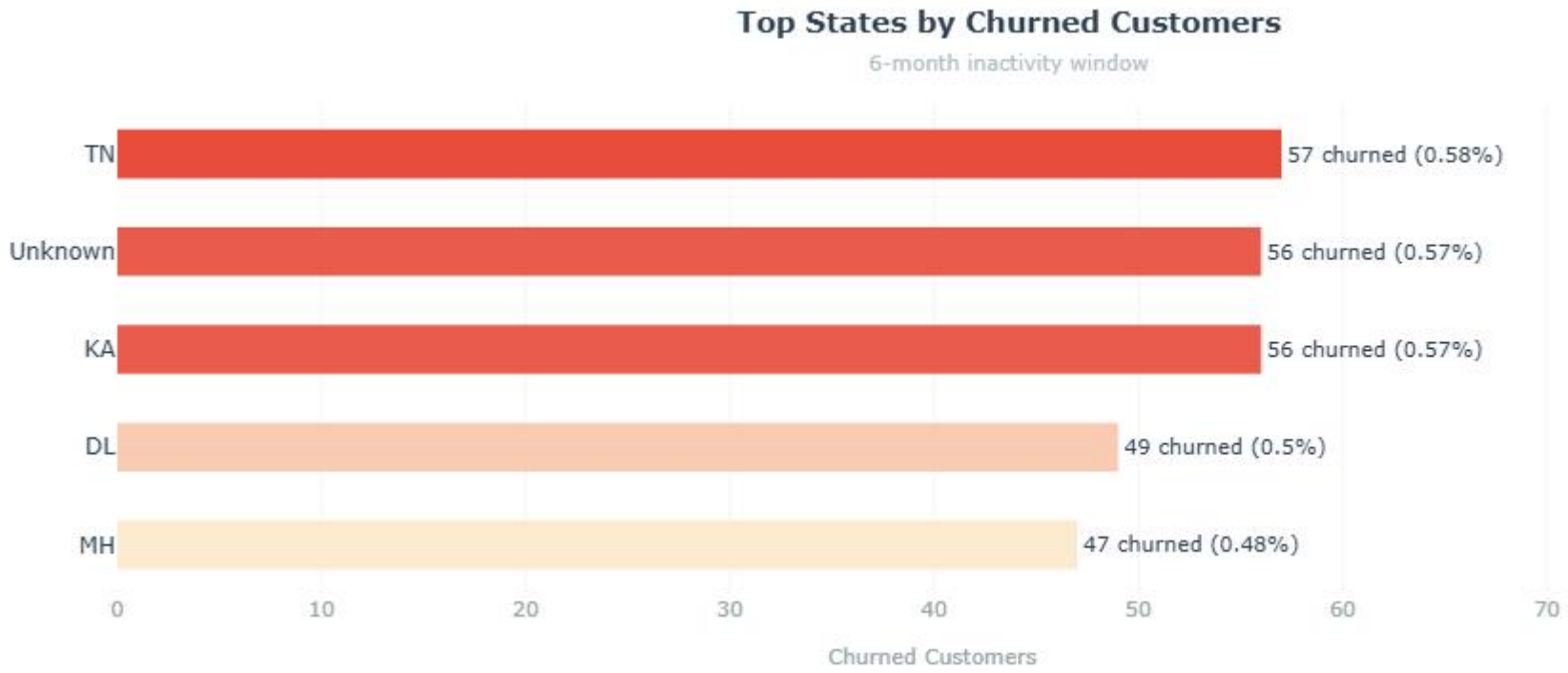
1. Maintain support and reliability across all payment methods.
2. Focus on engagement and retention rather than payment type.
3. Monitor customers with unknown payment methods to reduce potential risks.

11. Top states with highest churn rate:

- Tamil Nadu (TN): 9,771 customers, 57 churned, churn rate 0.58%.
- Karnataka (KA): 9,827 customers, 56 churned, churn rate 0.57%.
- Unknown: 9,744 customers, 56 churned, churn rate 0.57%.
- Delhi (DL): 9,787 customers, 49 churned, churn rate 0.50%.
- Maharashtra (MH): 9,871 customers, 47 churned, churn rate 0.48%.
- Churn rates are low across all states, with TN and KA showing slightly higher churn.

Actionable Insights:

1. Focus retention campaigns in TN and KA to reduce churn.
2. Investigate state-specific factors that may influence churn (e.g., service, logistics, offers).
3. Monitor “Unknown” customer locations for potential data quality or engagement issues.


Top States by Churned Customers
6-month inactivity window
TN
Unknown
KA
DL
MH
57 churned (0.58%)
56 churned (0.57%)
56 churned (0.57%)
49 churned (0.5%)
47 churned (0.48%)
0
10
20
30
40
50
60
70
Churned Customers

Key Insights (Overall Project)

- Total customers: 49,000, with very low overall churn (0.54%).
- Customer engagement is strong: average 20.41 orders per customer, average order value ₹2,547, and average customer spending ₹51,990.
- High-value customers dominate the customer base, with minimal churn (48,735 active vs 265 churned).
- Monthly churn shows a temporary spike in May 2023, otherwise stable.
- RFM analysis identifies 29,117 high-value customers, 19,618 at-risk, and 265 lost customers.
- Review scores have minimal impact on churn; other factors (recency, frequency, engagement) are more significant.
- Churn is consistently low across payment methods; slight differences exist for “Unknown” and UPI.
- Geographic analysis shows TN and KA with slightly higher churn rates, though overall churn remains low.

Actionable Insights (Overall)

1. Focus on retaining high-value and at-risk customers with loyalty programs and personalized engagement.
2. Monitor monthly churn trends to address temporary spikes proactively.
3. Implement targeted re-engagement campaigns for at-risk and lost customers.
4. Maintain consistent customer experience across states and payment methods.
5. Collect additional data on “Unknown” locations and monitor engagement signals for potential risks.

Recommendations

- **High-Value Customers:** Maintain engagement through loyalty programs, personalized offers, and proactive communication.
- **At-Risk Customers:** Implement re-engagement campaigns, reminders, and targeted incentives to reduce potential churn.
- **Lost Customers:** Launch win-back initiatives and analyze reasons for inactivity.
- **Monthly Churn Spikes:** Investigate operational or seasonal factors causing temporary churn spikes and address them proactively.
- **Geographic Focus:** Monitor TN and KA closely and tailor state-specific retention strategies.
- **Payment Methods:** Ensure seamless experience across all payment methods; focus on reducing friction for UPI and Unknown categories.
- **Customer Experience:** Collect feedback, monitor engagement, and improve touchpoints to prevent churn driven by indirect factors.

Conclusion

- Overall customer retention is strong, with very low churn (0.54%) and high engagement.

- High-value customers drive the majority of revenue, emphasizing the need for targeted retention strategies.

- RFM segmentation highlights opportunities to re-engage at-risk customers and recover lost customers.
- Data-driven insights allow the company to focus resources efficiently, enhance customer loyalty, and sustain revenue growth.

- Continuous monitoring of churn trends, customer behavior, and engagement metrics is critical to maintaining low churn and maximizing customer lifetime value.